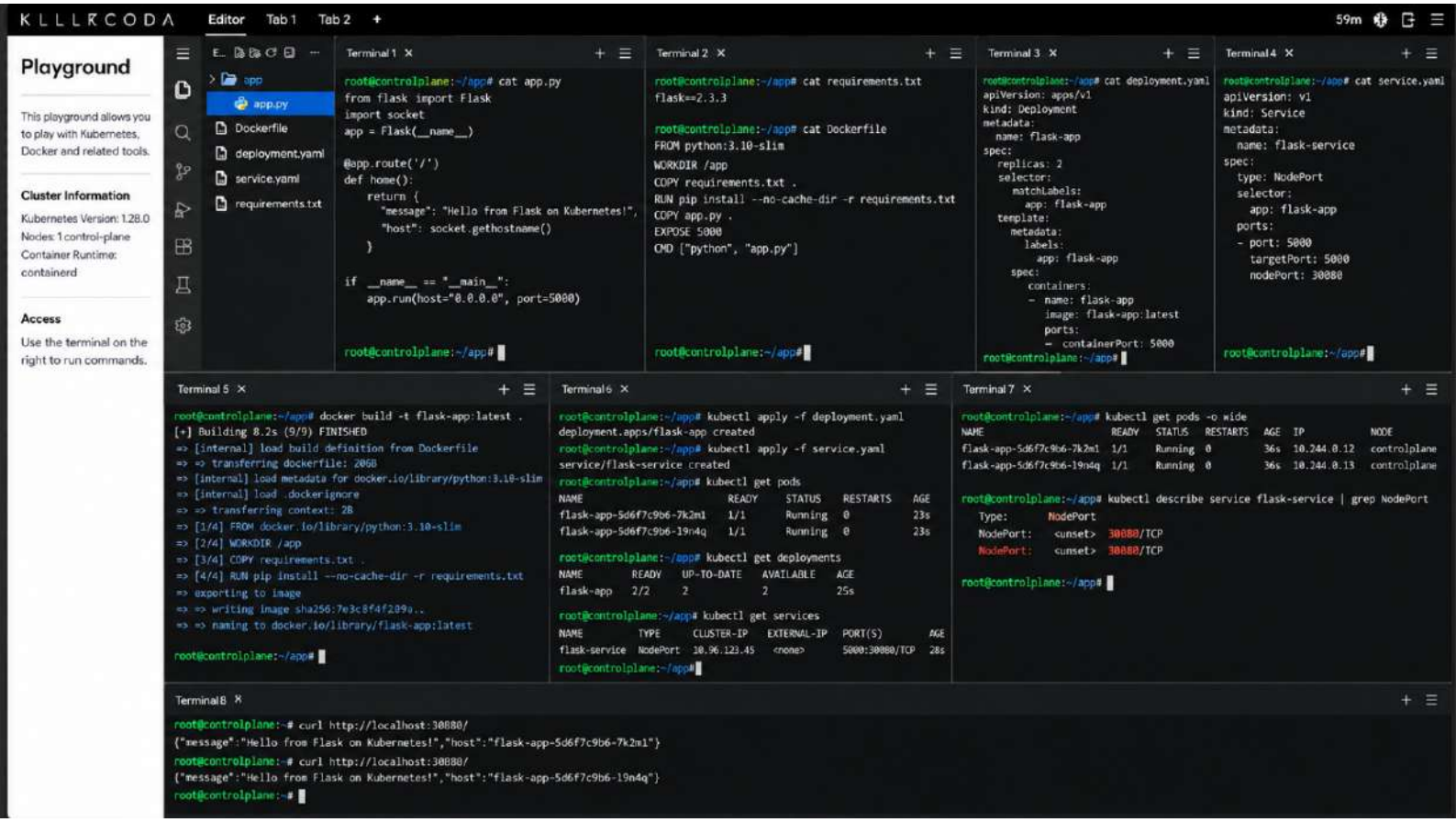

KLLLRCODA
Editor
Tab 1
Tab 2
59m
Playground
This playground allows you to play with Kubernetes, Docker and related tools.
Cluster Information
Kubernetes Version: 1.28.0
Nodes: 1 control-plane
Container Runtime: containerd
Access
Use the terminal on the right to run commands.
app
app.py
Dockerfile
deployment.yaml
service.yaml
requirements.txt
Terminal 1
root@controlplane:~/app# cat app.py
from flask import Flask
import socket
app = Flask(__name__)
@app.route('/')
def home():
return {
"message": "Hello from Flask on Kubernetes!",
"host": socket.gethostname()
}
if __name__ == "__main__":
app.run(host="0.0.0.0", port=5000)
root@controlplane:~/app#
Terminal 2
root@controlplane:~/app# cat requirements.txt
flask==2.3.3
root@controlplane:~/app# cat Dockerfile
FROM python:3.10-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
EXPOSE 5000
CMD ["python", "app.py"]
root@controlplane:~/app#
Terminal 3
root@controlplane:~/app# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
name: flask-app
spec:
replicas: 2
selector:
matchLabels:
app: flask-app
template:
metadata:
labels:
app: flask-app
spec:
containers:
- name: flask-app
image: flask-app:latest
ports:
- containerPort: 5000
root@controlplane:~/app#
Terminal 4
root@controlplane:~/app# cat service.yaml
apiVersion: v1
kind: Service
metadata:
name: flask-service
spec:
type: NodePort
selector:
app: flask-app
ports:
- port: 5000
targetPort: 5000
nodePort: 30080
root@controlplane:~/app#
Terminal 5
root@controlplane:~/app# docker build -t flask-app:latest .
[+] Building 8.2s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 206B
=> [internal] load metadata for docker.io/library/python:3.10-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.10-slim
=> [2/4] WORKDIR /app
=> [3/4] COPY requirements.txt .
=> [4/4] RUN pip install --no-cache-dir -r requirements.txt
=> exporting to image
=> => writing image sha256:7e3c8f4f209a..
=> => naming to docker.io/library/flask-app:latest
root@controlplane:~/app#
Terminal 6
root@controlplane:~/app# kubectl apply -f deployment.yaml
deployment.apps/flask-app created
root@controlplane:~/app# kubectl apply -f service.yaml
service/flask-service created
root@controlplane:~/app# kubectl get pods
NAME READY STATUS RESTARTS AGE
flask-app-5d6f7c9b6-7k2m1 1/1 Running 0 23s
flask-app-5d6f7c9b6-19n4q 1/1 Running 0 23s
root@controlplane:~/app# kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
flask-app 2/2 2 2 25s
root@controlplane:~/app# kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
flask-service NodePort 10.96.123.45 <none> 5000:30080/TCP 28s
root@controlplane:~/app#
Terminal 7
root@controlplane:~/app# kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE
flask-app-5d6f7c9b6-7k2m1 1/1 Running 0 36s 10.244.0.12 controlplane
flask-app-5d6f7c9b6-19n4q 1/1 Running 0 36s 10.244.0.13 controlplane
root@controlplane:~/app# kubectl describe service flask-service | grep NodePort
Type: NodePort
NodePort: <unset> 30080/TCP
NodePort: <unset> 30080/TCP
root@controlplane:~/app#
Terminal 8
root@controlplane:~# curl http://localhost:30080/
{"message":"Hello from Flask on Kubernetes!","host":"flask-app-5d6f7c9b6-7k2m1"}
root@controlplane:~# curl http://localhost:30080/
{"message":"Hello from Flask on Kubernetes!","host":"flask-app-5d6f7c9b6-19n4q"}
root@controlplane:~#